

GAP-10I Pairing Audit: No-New-Field Reduction and the Concurrent-Vector Obstruction

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Abstract

The generic two-sided UBT derivative $D_\mu X = \partial_\mu X + A_\mu X - XB_\mu$ appears to introduce two new connection fields. This note proves that this is not necessary in the pure Lorentz branch: preservation of the intrinsic Lorentz real slice and its metric reduces the pair uniquely, modulo a central term that cancels from the action, to the single existing spin connection, $D_\mu X = \partial_\mu X + \Omega_\mu X + X\Omega_\mu^\dagger$. However, the same reduction exposes a decisive obstruction. If $E_\mu = D_\mu \Theta / \sqrt{\mathcal{N}_0}$ is a nondegenerate torsion-free tetrad, then a Lorentz-real projection of Θ defines a concurrent spacetime vector V satisfying $\nabla_\mu V^\nu = \delta_\mu^\nu$. Hence the metric admits a proper homothety and $R^\rho_{\sigma\mu\nu} V^\sigma = 0$. Schwarzschild with nonzero mass, and generic Einstein geometries, do not obey this restriction. Thus the no-new-field Lorentz pairing is rigorously closed as a kinematic reduction but closed as a no-go for generic curved GR. A viable two-sided completion must derive a nontrivial relative bimodule action from the canonical UBT action without adding propagating degrees of freedom.

1 Algebraic conventions

Let $\mathbb{B} \simeq \text{Mat}(2, \mathbb{C})$ and let $\dagger$ denote Hermitian conjugation (complex conjugation followed by quaternion conjugation). The classical Lorentz slice is

$$W_L = \{X \in \mathbb{B} \mid X^\dagger = -X\} = \{ix^0 \mathbf{1} + x^k \mathbf{e}_k \mid x^a \in \mathbb{R}\}. \quad (1)$$

Equivalently, W_L is the fixed set of $\mathcal{J}X = -X^\dagger$.

Consider the most general regular bimodule connection on $\mathbb{B}$,

$$D_\mu X = \partial_\mu X + A_\mu X - XB_\mu. \quad (2)$$

2 Exact reduction of the pair

Theorem 2.1 (Lorentz-slice preserving pairs). *At a fixed spacetime point, Eq. (2) maps W_L into W_L for every $X \in W_L$ if and only if*

$$\boxed{B_\mu = -A_\mu^\dagger + c_\mu \mathbf{1}, \quad c_\mu \in \mathbb{R}.} \quad (3)$$

Proof. For $X^\dagger = -X$, slice preservation is equivalent to $(D_\mu X)^\dagger = -D_\mu X$. The connection terms give

$$(A_\mu + B_\mu^\dagger)X = X(A_\mu^\dagger + B_\mu). \quad (4)$$

Put $P_\mu = A_\mu + B_\mu^\dagger$. Taking $X = i\mathbf{1}$ in Eq. (4) gives $P_\mu = P_\mu^\dagger$. Taking $X = \mathbf{e}_k$ then gives $[P_\mu, \mathbf{e}_k] = 0$ for all three quaternion units. Their commutant in $\text{Mat}(2, \mathbb{C})$ is the center, so $P_\mu = c_\mu \mathbf{1}$; Hermiticity makes c_μ real. This is Eq. (3). The converse follows by direct substitution. $\square$

Theorem 2.2 (Metric-compatible reduction to one spin connection). *Assume in addition that the induced transport preserves the Lorentz quadratic form. Write*

$$A_\mu = \Omega_\mu + \alpha_\mu \mathbf{1}, \quad \text{tr } \Omega_\mu = 0. \quad (5)$$

Then metric compatibility requires $c_\mu = 2 \text{Re } \alpha_\mu$, and Eq. (2) reduces exactly to

$$\boxed{D_\mu X = \partial_\mu X + \Omega_\mu X + X \Omega_\mu^\dagger.} \quad (6)$$

Equivalently one may choose the representative

$$\boxed{A_\mu = \Omega_\mu, \quad B_\mu = -\Omega_\mu^\dagger.} \quad (7)$$

The common central one-form is invisible because it cancels between left and right multiplication.

Proof. Using Eq. (3), the connection part becomes

$$A_\mu X - X B_\mu = \Omega_\mu X + X \Omega_\mu^\dagger + (2 \text{Re } \alpha_\mu - c_\mu) X. \quad (8)$$

The final real scalar term is a local dilation of every Lorentz vector and therefore rescales the Lorentz bilinear form. Metric compatibility sets its coefficient to zero. Then $B_\mu = -\Omega_\mu^\dagger + \alpha_\mu \mathbf{1}$, and the same central $\alpha_\mu \mathbf{1}$ occurs on both sides of Eq. (2), so it cancels identically. The traceless spin lift fixes the representative Eq. (7). $\square$

Corollary 2.3 (No additional Lorentz connection fields). *In the pure gravitational Lorentz branch, A_μ and B_μ are not two independent fields. They are the left and right actions of the one spin connection*

$$\Omega_\mu = \frac{1}{4} \omega_{\mu ab} \Sigma^{ab}. \quad (9)$$

For specified tetrad and torsion, $\Omega = \Omega(e, T)$ is already fixed. In the torsion-free branch it is the Levi-Civita spin connection, and in Cartesian inertial Minkowski gauge $A_\mu = B_\mu = \Omega_\mu = 0$.

3 The obstruction exposed by the reduction

Theorem 3.1 (Concurrent-vector theorem). *Let the torsion-free metric-compatible derivative be Eq. (6). Suppose*

$$D_\mu \Theta = s E_\mu, \quad s := \sqrt{\mathcal{N}_0}, \quad E_\mu \in W_L, \quad (10)$$

and E_μ is a nondegenerate tetrad. Then there exists a real spacetime vector field V satisfying

$$\boxed{\nabla_\mu V^\nu = \delta_\mu^\nu.} \quad (11)$$

Consequently

$$\mathcal{L}_V g_{\mu\nu} = 2g_{\mu\nu}, \quad R^\rho{}_{\sigma\mu\nu} V^\sigma = 0. \quad (12)$$

Proof. Equation (6) commutes with $\mathcal{J}X = -X^\dagger$. Therefore the Lorentz-real projection

$$X := \frac{1}{2}(\Theta + \mathcal{J}\Theta) \in W_L \quad (13)$$

obeys $D_\mu X = sE_\mu$, because $\mathcal{J}E_\mu = E_\mu$. Write $X = X^a \mathbf{u}_a$ and $E_\mu = e_\mu^a \mathbf{u}_a$ in a Lorentz basis and define

$$V^\nu := s^{-1} e_a^\nu X^a. \quad (14)$$

Using the inverse tetrad postulate and $D_\mu X^a = s e_\mu^a$ gives

$$\nabla_\mu V^\nu = s^{-1} e_a^\nu D_\mu X^a = e_a^\nu e_\mu^a = \delta_\mu^\nu. \quad (15)$$

Symmetrization yields the homothety equation. Antisymmetrizing two covariant derivatives of V yields the curvature-kernel identity. $\square$

Proposition 3.2 (Schwarzschild exclusion). *The Schwarzschild metric with mass $M \neq 0$ admits no vector satisfying Eq. (11); indeed it admits no proper homothety $\mathcal{L}_V g = 2g$.*

Proof. Average a putative homothetic vector over the rotational isometry group. The averaged vector remains homothetic and has the form $V = T(t, r)\partial_t + R(t, r)\partial_r$. For

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2, \quad f(r) = 1 - \frac{2M}{r}, \quad (16)$$

the angular component of $\mathcal{L}_V g = 2g$ gives $R = r$. The rr component then gives

$$-\frac{rf'(r)}{f(r)^2} + \frac{2}{f(r)} = \frac{2}{f(r)}, \quad (17)$$

so $f'(r) = 0$, contradicting $f'(r) = 2M/r^2 \neq 0$. $\square$

Remark 3.3 (Why constants in a potential do not remove the obstruction). A central constant or central one-form cancels from the bimodule derivative and cannot change Eq. (11). A potential may make an additional field auxiliary or fix its vacuum value, but a constant parameter cannot provide the spacetime-dependent relative left/right action required to evade the concurrent-vector restriction. Any such relative component must be derived algebraically from the canonical action and shown to carry no new propagating degree of freedom.

4 Defensible gap status

New exact subclosures

GAP-10I-PAIR-KIN: CLOSED [L1]. Lorentz-slice and metric compatibility reduce the apparent fields (A_μ, B_μ) to the single existing spin connection, modulo a central term that cancels identically. In flat inertial gauge both vanish.

GAP-10I-PAIR-GR: CLOSED AS NO-GO [L1]. That pure Lorentz pairing makes a nondegenerate torsion-free tetrad generated by one Θ imply a concurrent vector and a proper homothety. It therefore cannot represent generic curved GR and excludes Schwarzschild with $M \neq 0$.

GAP-10I-2S / GAP-10I-CURVED: NARROWED. The remaining admissible route must derive a nontrivial relative bimodule connection from the canonical action, prove that it is composite or auxiliary rather than propagating, and then establish self-consistent existence and regularity. Merely naming two independent connections is not a closure.

GAP-10D: NOT CLOSED BY THIS REDUCTION. The result removes a false no-new-field shortcut and identifies the exact missing object in the variational bridge. The Palatini/Lovelock endpoint remains conditional until the relative action, its algebraic elimination, and the induced metric variation are derived from canonical UBT.